Nondistortionary elicitation of beliefs

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Introduction

- Many experiments involve incentivized truthful reporting of beliefs
  - examples: self-confidence, cognitive uncertainty, information acquisition, dynamic games,
  - incentives improve accuracy (Schlag et al. (2015) and many others).
- But combined with actions, incentivization may distort behavior:
  - *action-dependent* questions,
  - Example: questions about relative class rank, future choices.
- Distortions are problematic:
  - they affect interpretation of experimental results,
  - make honest and quiet instructions (Danz et al. (2022)) difficult,
  - ethical and legal issues.
- Typical solutions may introduce other problems:
  - don't tell the subject,
  - don't incentivize,
  - assume narrow-bracketing,
  - ask (many) *action-independent* questions.

Which questions can be incentivized without distorting behavior?

- Pęski and Stewart (2025) characterize all incentivizable questions in **static problems**:
  - Incentivizable questions must be "aligned" with payoffs.
- Pęski and Stewart (2026): **Dynamic problems** have new issues:
  - richer learning environment (Chambers and Lambert (2021)),
  - multiple points of asking questions, and
  - multiple possible points of distortions.
- Different ways actions can affect arrival of new information:
  - action-independent,
  - actions affect the quality of information,
  - actions affect the distribution of states.

Literature

Empirical: static problems

- "hedging": Blanco et al. (2010), Danz et al. (2022)
- beliefs in games: Palfrey and Wang (2009), Armantier and Treich (2009)
- confidence, cognitive uncertainty: Enke and Graeber (2023), Exley and Nielsen (2024)
- methods: Schlag et al. (2015), Healy and Leo (2024)

Empirical: dynamic problems

- learning, motivated beliefs: Möbius et al. (2022), Zimmermann (2020)
- information provision: Haaland et al. (2023)
- field, job search: Abebe et al. (2020), Carranza et al. (2022)

Theory

- scoring rules: Becker et al. (1964), Hossain and Okui (2013), Lambert et al. (2008)
- incentives in experiments, "contamination": Azrieli et al. (2018), Healy (2024)
- Chambers and Lambert (2021): dynamic elicitation, "moral hazard", action-independent information
- information acquisition: Tsakas (2020)
- static elicitation with many questions: Chen et al. (2026).

Mechanism design with a pre-existing mechanism

- Calzolari and Pavan (2006), Philippon and Skreta (2012), Tirole (2012), Dworczak (2020)

Static problems

IQ test.

Example: IQ test

- Subject writes an IQ test with two True-False problems.
- Each correct answer gives one point; the subject maximizes the expected score.
- Beliefs about the correct answers are privately known.

- $\Theta = A = \{T_1, F_1\} \times \{T_2, F_2\}$
- score:

$$u(a, \theta) = 1\{a_1 = \theta_1\} + 1\{a_2 = \theta_2\}$$

- belief $p \in \Delta\Theta$.
  Belief allows for correlation between θ_1 and θ_2.

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"What is your expected score?"

"What is the probability that both of your answers are correct?"

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Static problems

- Decision problem (A, u)
  - finite action set A,
  - finite state space Θ,
  - privately known belief $p \in \Delta\Theta$,
  - payoffs $u : A \times \Theta \longrightarrow \mathbb{R}$.
- Best responses

$$\mathbb{B}^u(p) = \arg\max_{a \in A} \mathbb{E}_p u(a, \theta)$$

- Value function

$$U^*(p) = \max_{a \in A} \mathbb{E}_p u(a, \theta)$$

- $X : A \times \Theta \longrightarrow \mathbb{R}$ *action-dependent* question about the *state*.
  - DM is asked to report $r = \mathbb{E}_p X(a, \cdot)$
- Assumption: one question only!

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"What is your expected score?" is equivalent to

$$X(a, \theta) = u(a, \theta)$$

"What is your expected ex post regret from choosing action a?" is equivalent to

$$X(a, \theta) = \max_b u(b, \theta) - u(a, \theta)$$

"What is the probability of state θ_0?" is equivalent to

$$X(a, \theta) = 1(\theta = \theta_0)$$

- Asking $|\Theta| - 1$ action-independent questions is sufficient to elicit any belief. But here we ask only one (!) question.

"What is the probability that your action matches the state?"

$$X(a, \theta) = 1\{a = \theta\}$$

Incentivizable or not, depending on the size of the game, payoffs, and other details.

- Question $X(a,\theta)$ is *incentivizable* in static decision problem (A,u) if there exists nondistortionary expansion $(A\times\mathbb{R},v)$ such that for every $p\in\Delta\Theta$

$$\text{no distortions: } \{a:(a,r)\in\mathbb{B}^v(p)\}=\mathbb{B}^u(p),$$
$$\text{truthful reporting: } \mathbb{B}^v(p)\subseteq\{(a,r):r=\mathbb{E}_pX(a,\theta)\}$$

  - $r\in\mathbb{R}$ - report of the expectation of a variable $X(a,\theta)$
  - equivalently: $\mathbb{B}^v(p)=\{(a,\mathbb{E}_pX(a,\theta)):a\in\mathbb{B}^u(p)\}$
- In principle, $v(a,r,\theta)$ is independent of u, but typically it is related (benchmark: experimenter knows u and observes a and θ).

Theorem

The following questions are incentivizable with BDM (1964):

- $X(a, \theta) = u(a, \theta) + d(\theta)$,
- $X(a, \theta) = d(\theta)$.

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- X is equivalent to Y if there exists $\gamma : A \to \mathbb{R} \setminus \{0\}$ and $\kappa : A \to \mathbb{R}$ such that

$$Y(a,\theta) = \gamma(a)X(a,\theta) + \kappa(a).$$

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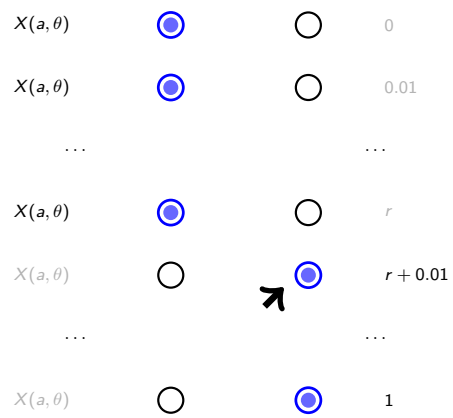
- W.l.o.g. normalize $0 < X(a,\theta) < 1$.
- subject reports $r \in [0,1]$,
- random number $x \sim U[0,1]$ is drawn uniformly,
- if $x \leq r$, the subject receives $X(a,\theta)$,
- otherwise, if $r \leq x$, the subject receives x.

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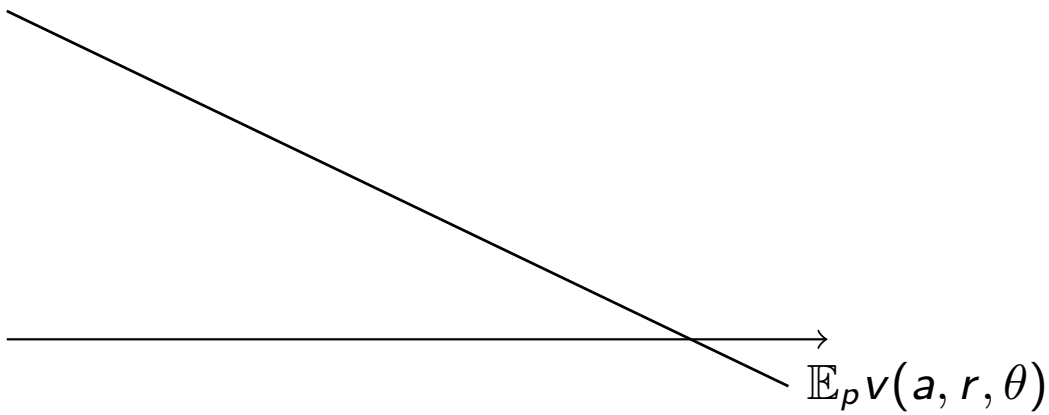
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$$\begin{aligned}
&\max_r \mathbb{E}_p V(r,a,.) \\
&= \max_r \left[\int_0^r \mathbb{E}_p X \, dx + \int_r^1 x \, dx \right] \\
&= \max_r \left[r\mathbb{E}_p X + \tfrac{1}{2}(1-r^2) \right] \\
&= \tfrac{1}{2}(\mathbb{E}_p X)^2 + \tfrac{1}{2} \\
&= \tfrac{1}{2}(\mathbb{E}_p u(a,.) + \mathbb{E}_p d)^2 + \tfrac{1}{2}.
\end{aligned}$$

Static problems

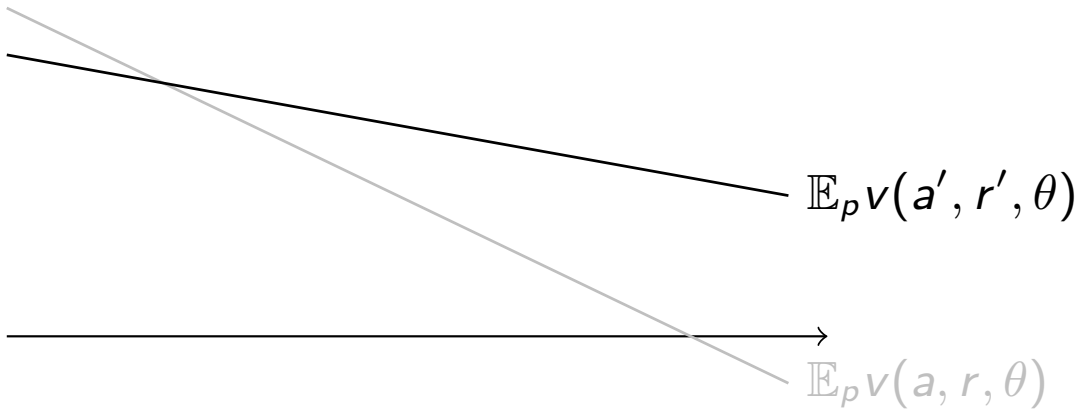
Value matching



- Value function $V^*(p) = \max_{a,r} \mathbb{E}_p v(a, r, \theta)$ is an upper envelope of affine functions of p: it is convex, and
  - it is strictly convex at p whenever two different best replies (a, r) attain the maximum at p,
  - it is affine on an interval where the best reply is constant,
  - it is smooth and strictly convex on an interval where the best reply changes continuously.

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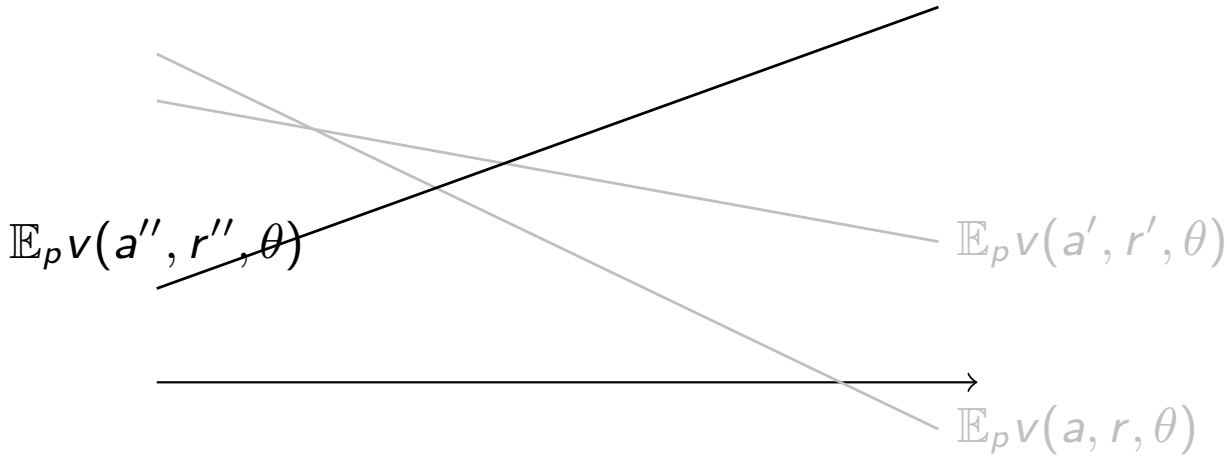
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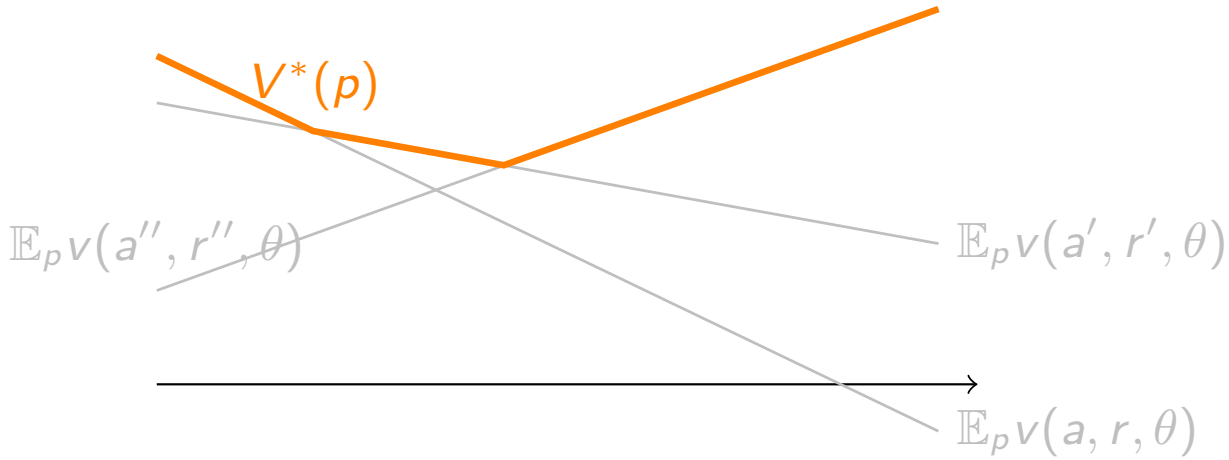
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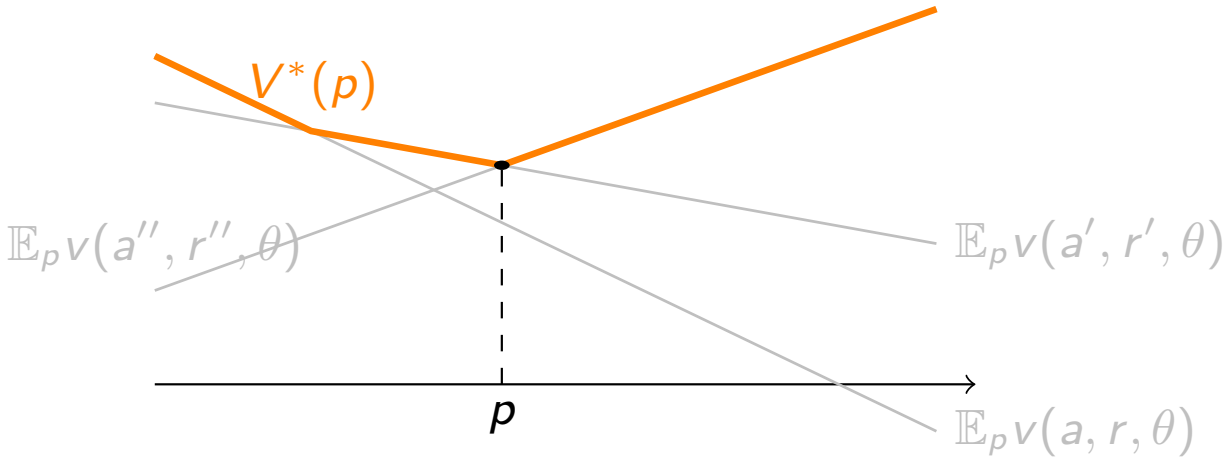
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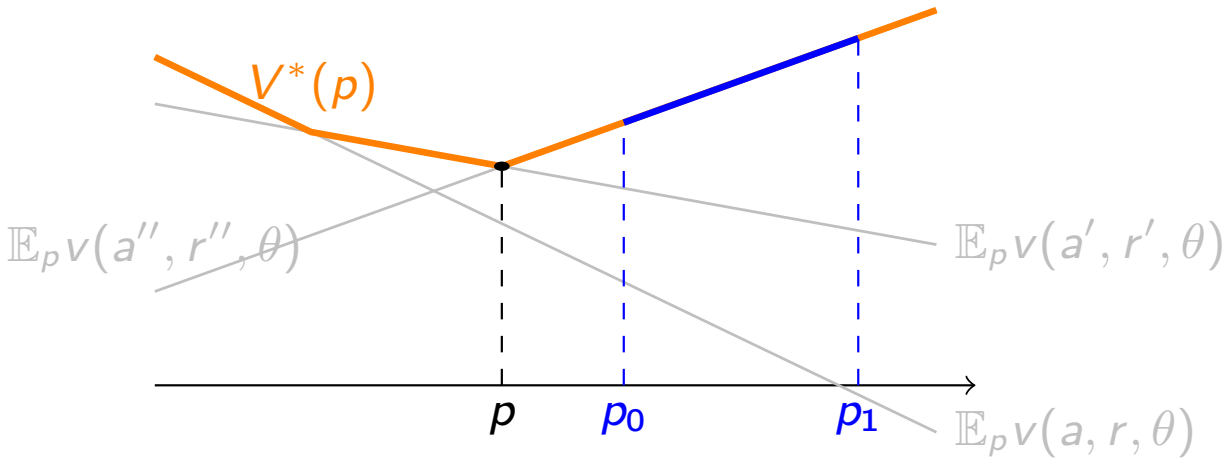
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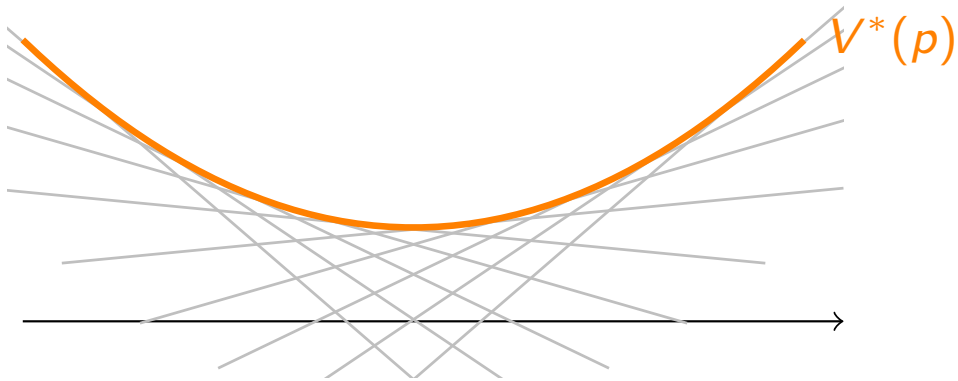
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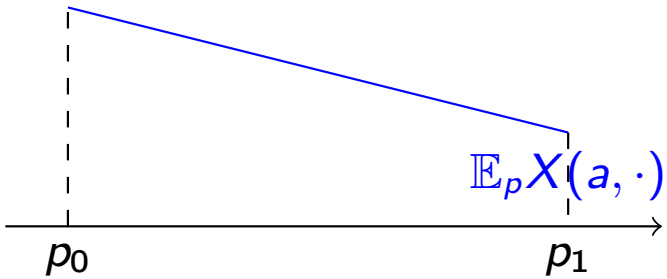
- If a and b are best responses at beliefs p_0 and p_1, they are best responses on the segment between them, and

$$\max_r \mathbb{E}_p v(a, r, \theta) = V^*(p) = \max_r \mathbb{E}_p v(b, r, \theta).$$

- Test: failure of incentivizability – beliefs p_0 and p_1 such that
  - (i) the DM is indifferent between a and b for $p \in [p_0, p_1]$,
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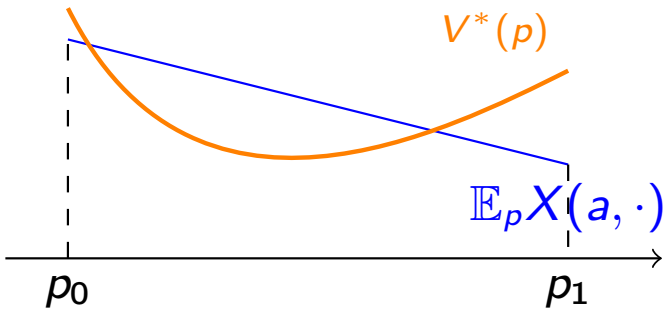
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changing reports $\implies$ str. convex value

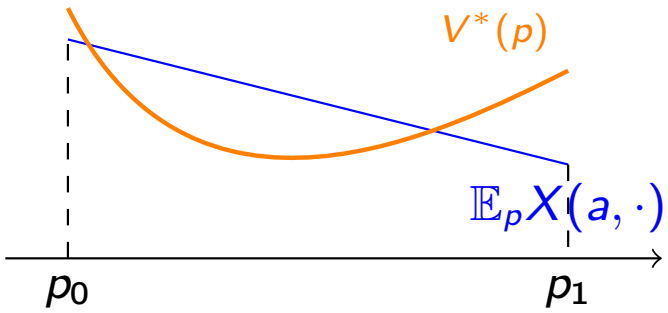
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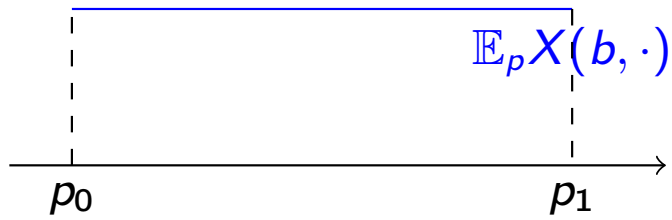
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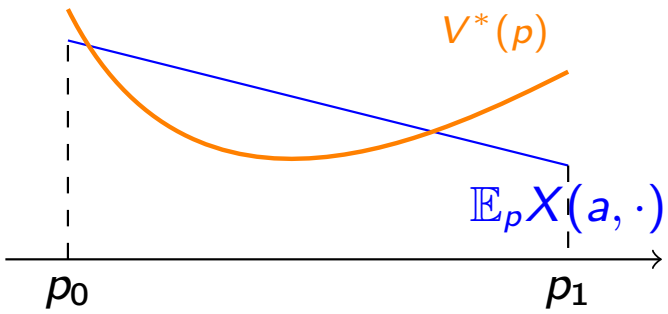


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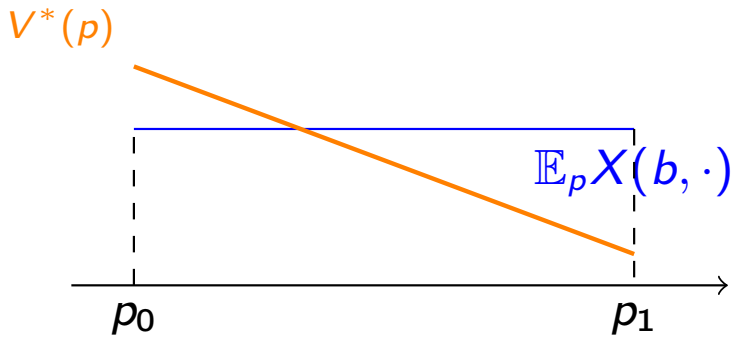
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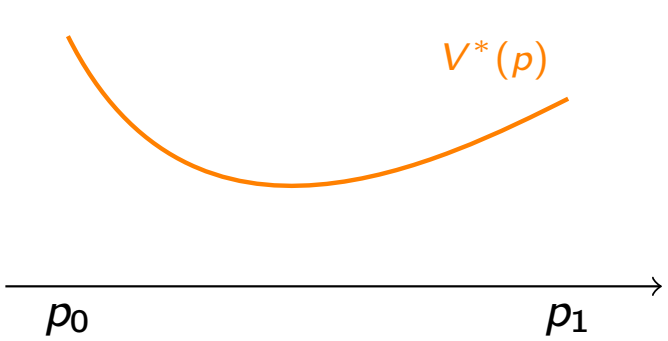
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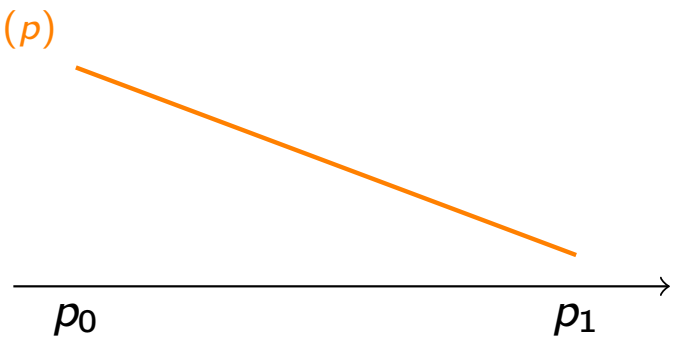
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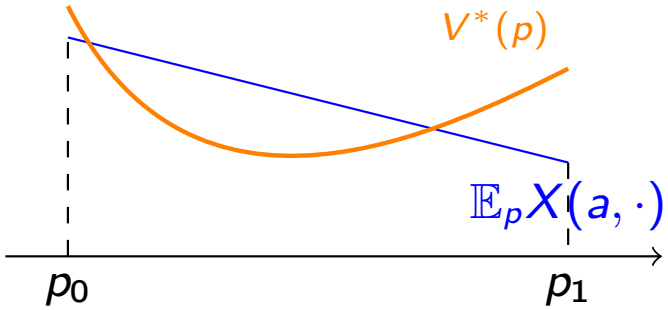


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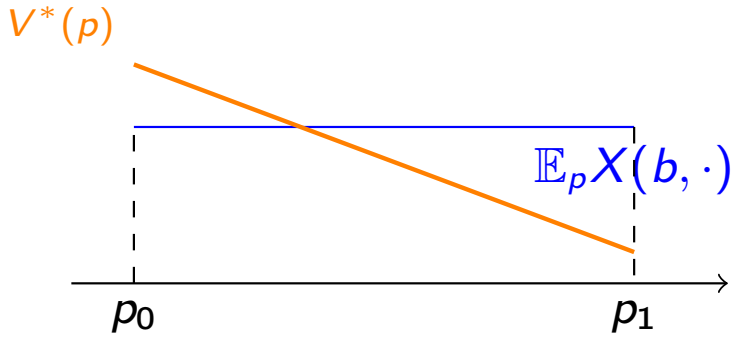
- Contradiction with incentivizability!

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Example: IQ test

Value matching

- "What is the probability that both of your answers are correct?" $X(a, \theta) = 1\{a = \theta\}$.
- Two beliefs over $\theta = (\theta_1, \theta_2)$ with $x, y, z \in (\frac{1}{4}, \frac{1}{2})$ and $y \neq z$:

	T_2	F_2
T_1	x	$\frac{1}{2} - x$
F_1	y	$\frac{1}{2} - y$

beliefs p_0

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beliefs p_1

- (i) The subject is indifferent between $a = (T_1, T_2)$ and $b = (F_1, T_2)$ for all $p \in [p_0, p_1]$.
- (ii) $a = (T_1, T_2)$: both answers correct iff $\theta = (T_1, T_2)$, so

$$\mathbb{E}_{p_0} X(a, \cdot) = x = x = \mathbb{E}_{p_1} X(a, \cdot).$$

- (iii) $b = (F_1, T_2)$: both answers correct iff $\theta = (F_1, T_2)$, so

$$\mathbb{E}_{p_0} X(b, \cdot) = y \neq z = \mathbb{E}_{p_1} X(b, \cdot).$$

- $\Longrightarrow$ X is not incentivizable.

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- $\Longrightarrow$ X is not incentivizable.

Example: IQ test

Value matching

- "What is the probability that both of your answers are correct?" $X(a, \theta) = 1\{a = \theta\}$.
- Two beliefs over $\theta = (\theta_1, \theta_2)$ with $x, y, z \in (\frac{1}{4}, \frac{1}{2})$ and $y \neq z$:

	T_2	F_2	
T_1	x	$\frac{1}{2} - x$	$\frac{1}{2}$
F_1	y	$\frac{1}{2} - y$	$\frac{1}{2}$
	$> \frac{1}{2}$		

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$=$

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Adjacency Lemma (Pęski and Stewart (2025))

If X is incentivizable, then, for each adjacent pair $\{a, b\}$, the restriction of X to $\{a, b\}$ is equivalent to

$$s\,u(a,\theta) + d(\theta) \quad \text{for } s \in \{0,1\} \text{ and } d \in \mathbb{R}^{\Theta}$$

(d and equivalence constants may depend on the pair).

- Proof: value matching, i.e., the test (i)–(iii) applied to each adjacent pair.
- The Lemma has extra bite on adjacency cycles.

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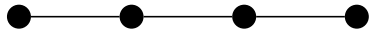
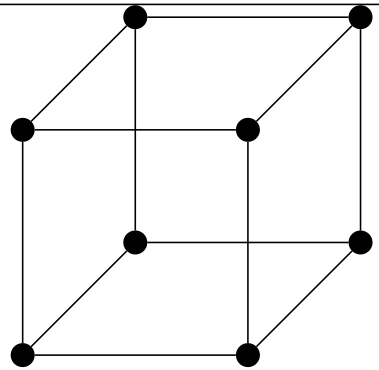
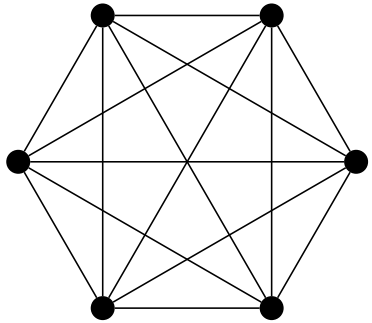
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Incentivizable questions in static problems

	Tree	**Product problem**	**Complete graph**
Adjacency graph			
Examples	monotone problems, cognitive uncertainty (Enke and Graeber, 2023),	random problem selection (Azrieli et al., 2018), test with ≥ 3 questions	multiple choice question, prediction problems
Representation (necessary = sufficient)	Adjacency Lemma on each adjacency pair	$\sum_i s_i u_i(a_i, \theta_i) + d(\theta)$ where $s_i \in \mathbb{R}$ and equivalents*	$s\, u(a, \theta) + d(\theta)$ where $s \in \{0, 1\}$ and equivalents*

***Additional independence assumptions on payoffs**

Theorem

Suppose that

- *the adjacency graph of the action set A is complete, and*
- *vectors $\{\bar{u}(x,\cdot) - \bar{u}(a,\cdot) : x = b, c, d\}$ are independent for any distinct a, b, c, d.*

Then, X is incentivizable in (A, u) if and only if it is equivalent to

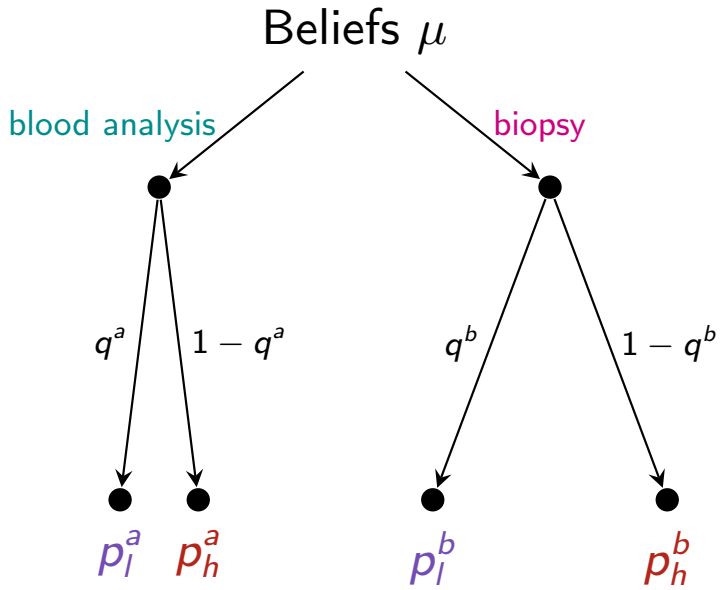
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Dynamic problems: Period 1

Doctor.

Example: Doctor

- Doctor chooses a method to diagnose tumor count:
  - blood analysis or biopsy,
  - biopsy improves information, but it is invasive.
- After observing the outcome, she chooses treatment:
  - active surveillance or chemotherapy
- The amount of learning is *privately* known to the doctor.
- Doctor maximizes QALY (and biopsy is costly).



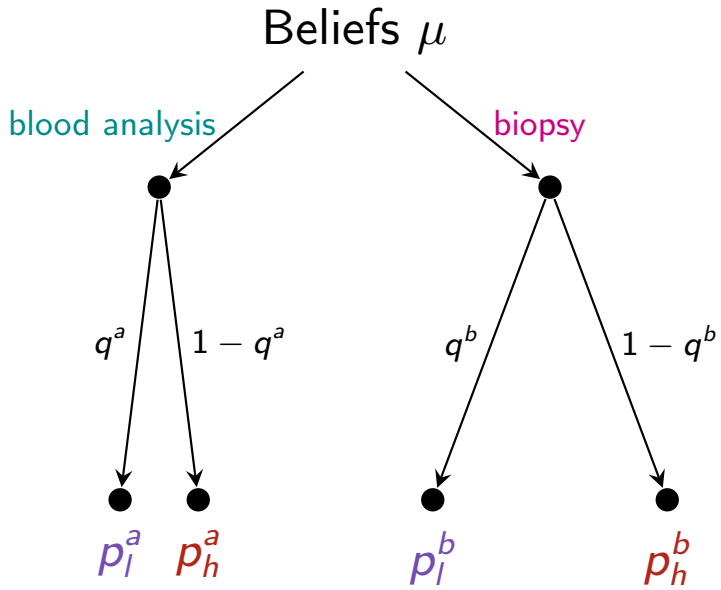
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"What is the expected QALY (i.e., your expected pay-offs)?"

"How likely are you to prescribe chemotherapy?"



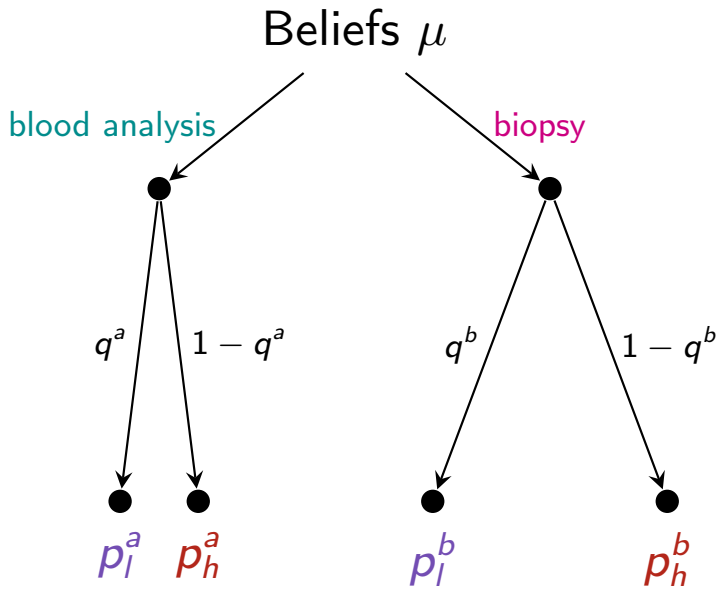
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Dynamic problems: Period 1

Decision problem

- Dynamic decision problem: (A_1, A_2, u)
  - A_t - finite action set in period t
  - $u : A_1 \times A_2 \times \Theta \longrightarrow \mathbb{R}$ payoffs
- Expected utility, Bayesian framework $\implies$ expected payoffs:

$$\text{Period 2: } u^*(a_1, p) = \max_{a_2} \mathbb{E}_p u(a_1, a_2, \theta) \text{ for } p \in \Delta\Theta$$

$$\text{Period 1: } U^*(\mu) = \max_{a_1} \mathbb{E}_{\mu(a_1)} u^*(a_1, p) \text{ for } \mu(a_1) \in \Delta\Delta\Theta$$

- Actions vs information
  - Information is not affected by actions: $\mu \in \Delta\Delta\Theta$
  - Actions affect the quality of information: $\mu : A_1 \longrightarrow \Delta\Delta\Theta$ and $\mathbb{E}_{\mu(a_1)} p = \mathbb{E}_{\mu(b_1)} p$,
  - Actions affect states.

- $X : A_1 \times \Delta\Theta \longrightarrow \mathbb{R}$ *action-dependent* period 1 question about *period 2 beliefs*.
  - "What are your expected payoffs?" is equivalent to
    $$X(a_1, p) = u^*(a_1, p)$$
  - "How likely will you believe that chemotherapy is optimal?"
    $$X(a_1, p) = 1\{p > p^*\}.$$
- DM is asked to report $r = \mathbb{E}_{\mu(a_1)} X(a_1, \cdot)$

Dynamic problems: Period 1

Incentivizability

- Question $X(a_1, p)$ is *incentivizable in period 1* in dynamic decision problem (A_1, A_2, u) if there exists nondistortionary expansion $(A_1 \times \mathbb{R} \times S_1, A_2 \times S_2, v)$ such that

$$\text{no distortions in } t = 1\text{: } \mathbb{B}_1^v(\mu) \subseteq \mathbb{B}_1^u(\mu) \times \mathbb{R} \times S_1,$$
$$\text{no distortions in } t = 2\text{: } \mathbb{B}_2^v(p; a_1, r, s_1) \subseteq \mathbb{B}_2^u(a_1, p) \times S_2$$
$$\text{truthful reporting: } \mathbb{B}_1^v(\mu) \subseteq \{(a_1, r, s_1) : r = \mathbb{E}_{\mu(a_1)} X(a_1, p)\}$$

  - $r \in \mathbb{R}$ - period 1 report of the expectation of a variable $X(a_1, p)$
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$$X(a_1, p) = u^*(a_1, p) + f(p)$$

- Becker et al. (1964): DM reports r, $x \sim U[0,1]$ is drawn; DM receives $X(a_1, p)$ if $x \leq r$, and x otherwise (w.l.o.g. $0 < X < 1$):

$$v(a_1, r, p) = r\, X(a_1, p) + \tfrac{1}{2}(1 - r^2).$$

- Problem: p is not observed!

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- Questions about *payoffs*: no further enhancement needed.
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Sufficient conditions: BDM + supporting actions

- But we still want to elicit a question about *learning*:

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- Inspired by Chambers and Lambert (2021): any continuous f can be approximated by "finitely continuous"

$$f(p) = \max_{d \in D_+} d(p) - \max_{d \in D_-} d(p)$$

  for finite $D_+, D_- \subseteq \mathbb{R}^\Theta$.
- Supporting actions in period 2: $S_2 = D_+ \times D_-$, and

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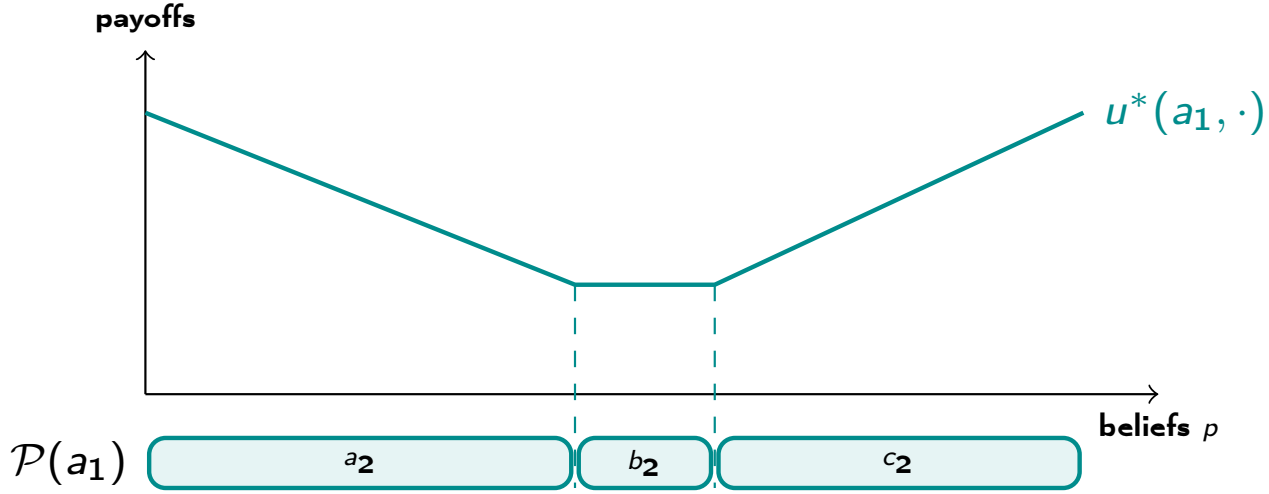
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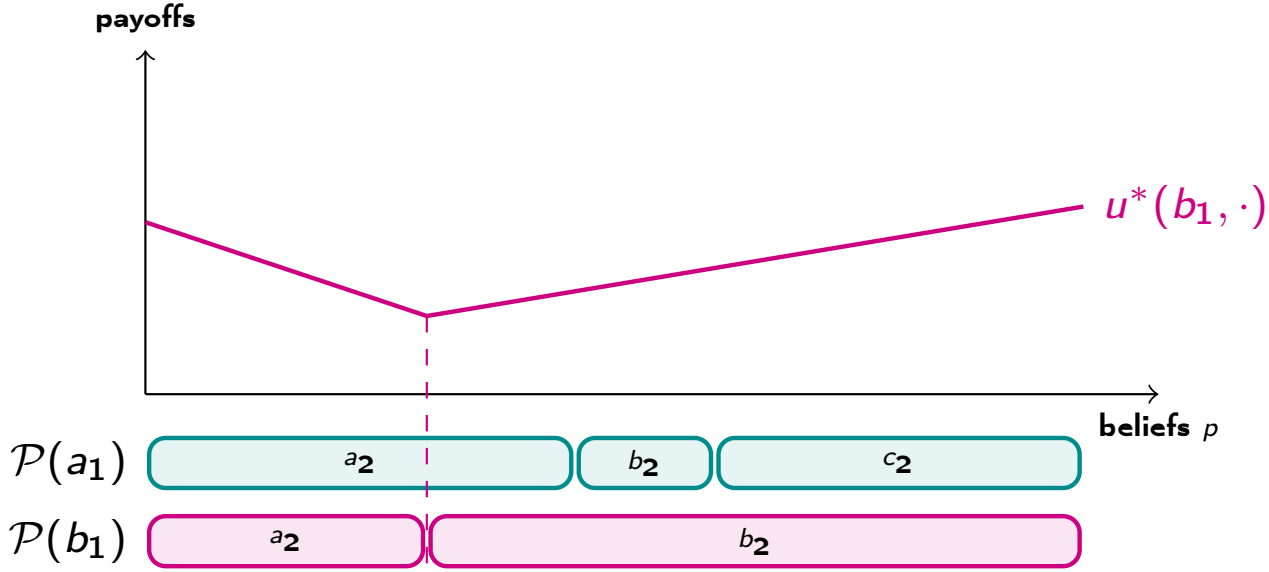
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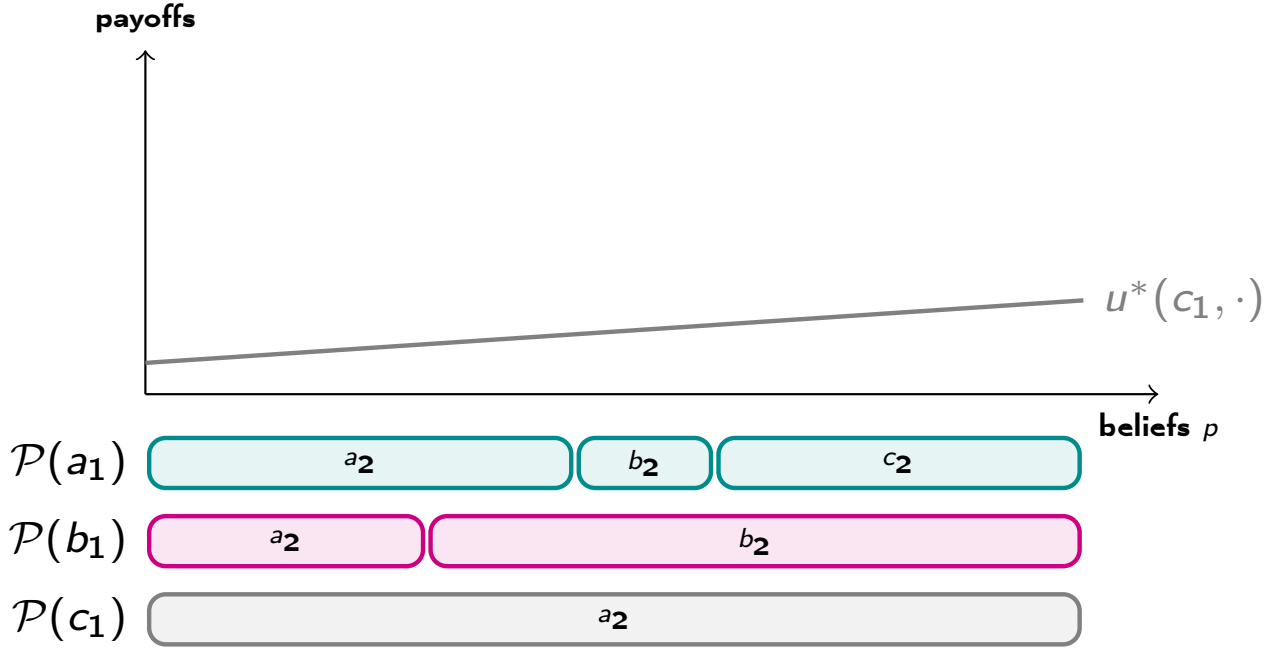


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Dynamic problems: Period 1

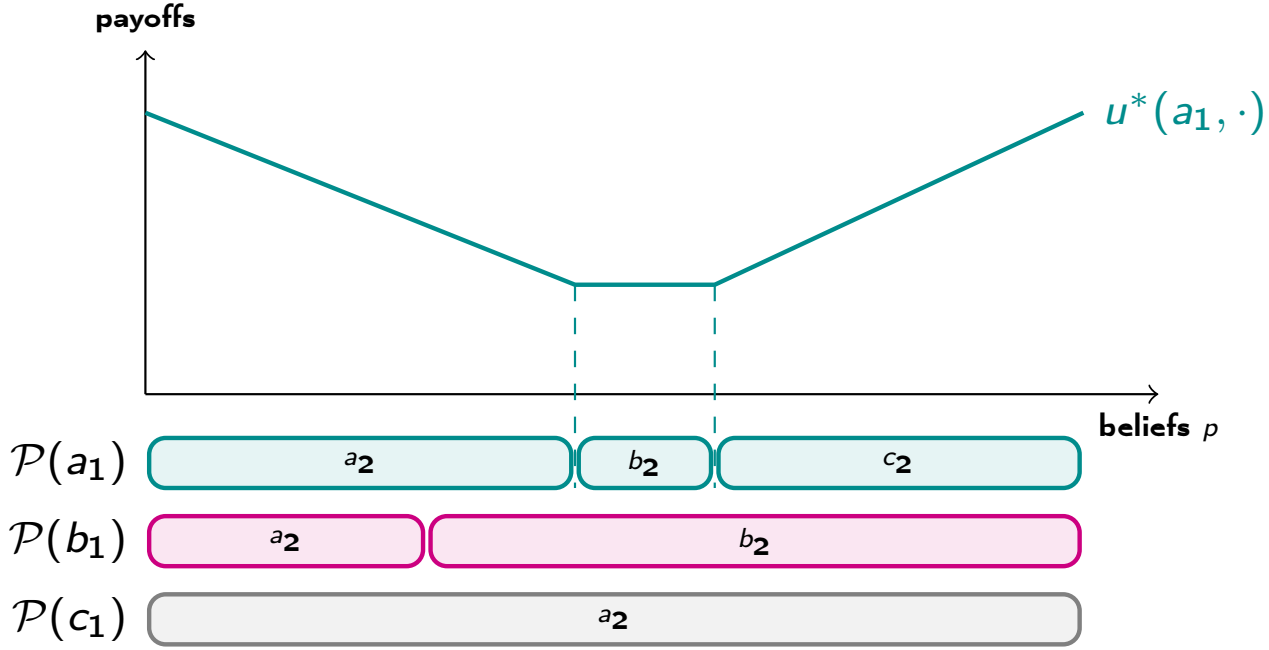
Belief partitions

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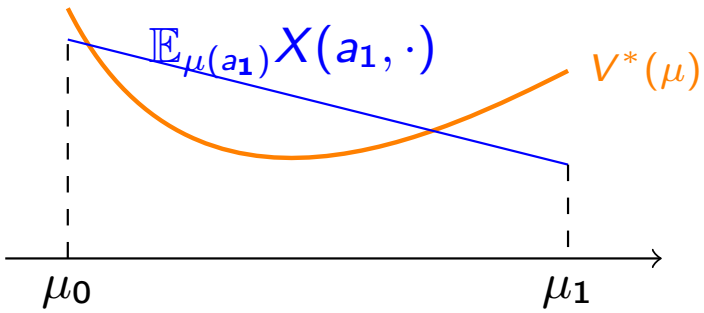


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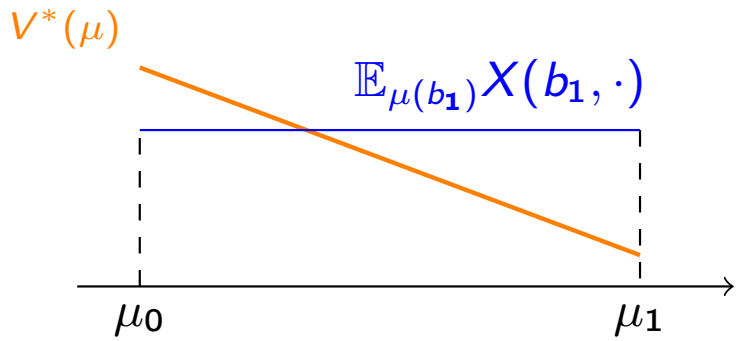
Example: Doctor

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changing beliefs $\Longrightarrow$ str. convex value



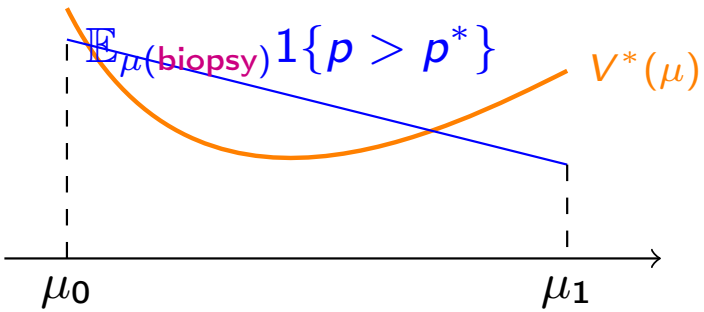
constant beliefs $\Longrightarrow$ affine value

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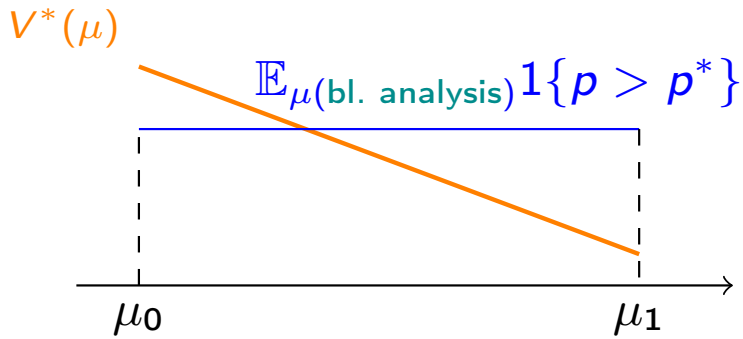
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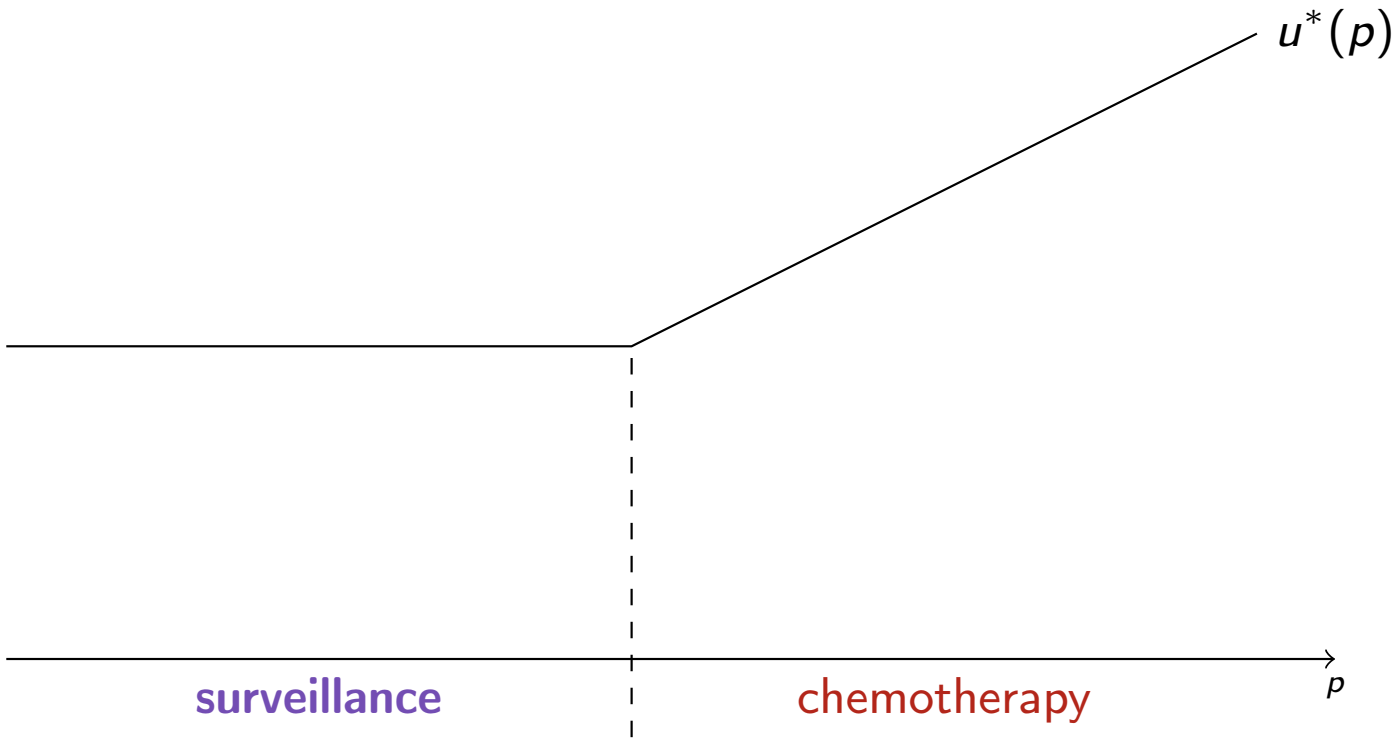


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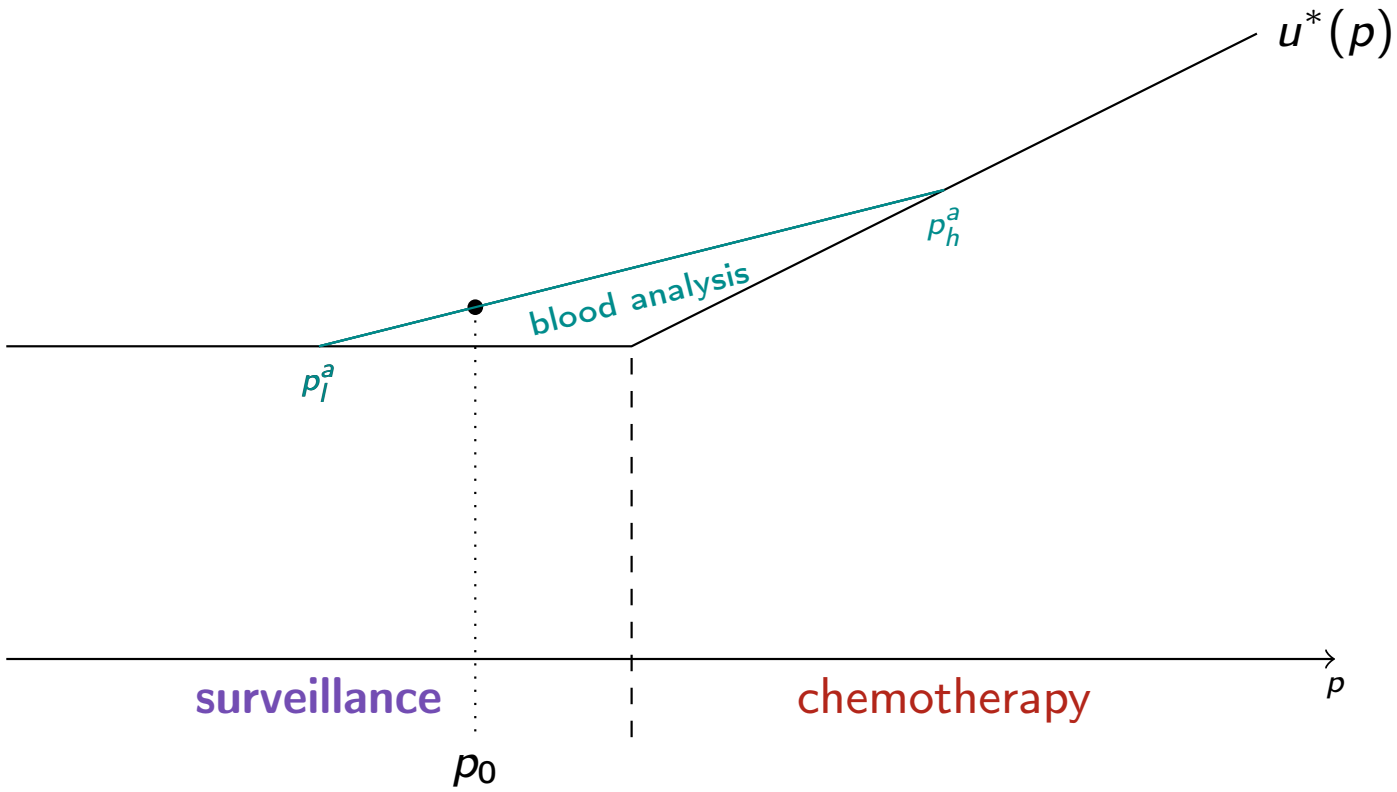
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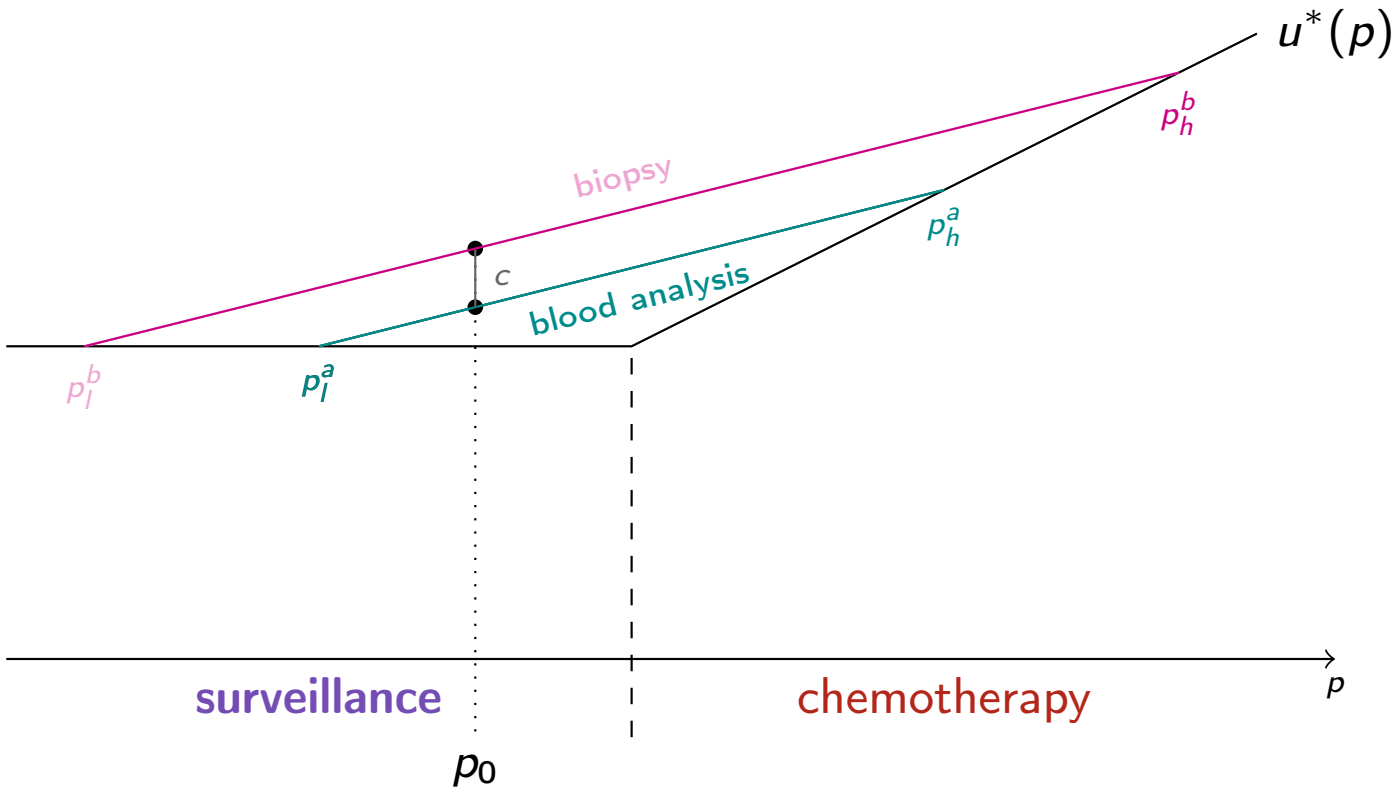
Value matching



- There are beliefs μ_0 and μ_1 that keep the indifference between the two diagnostic choices, and keep the probability of chemotherapy
  - constant after blood analysis, and
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Example: Doctor

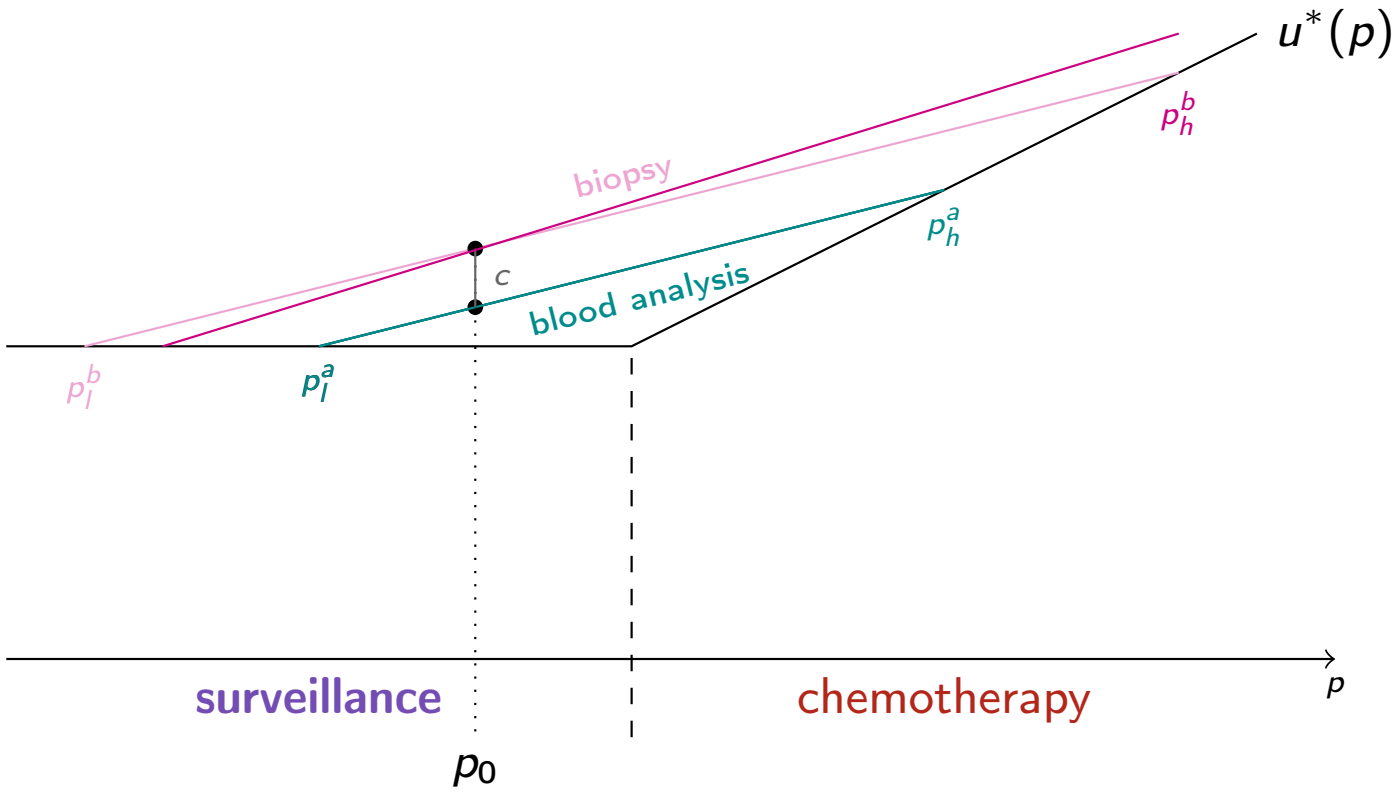
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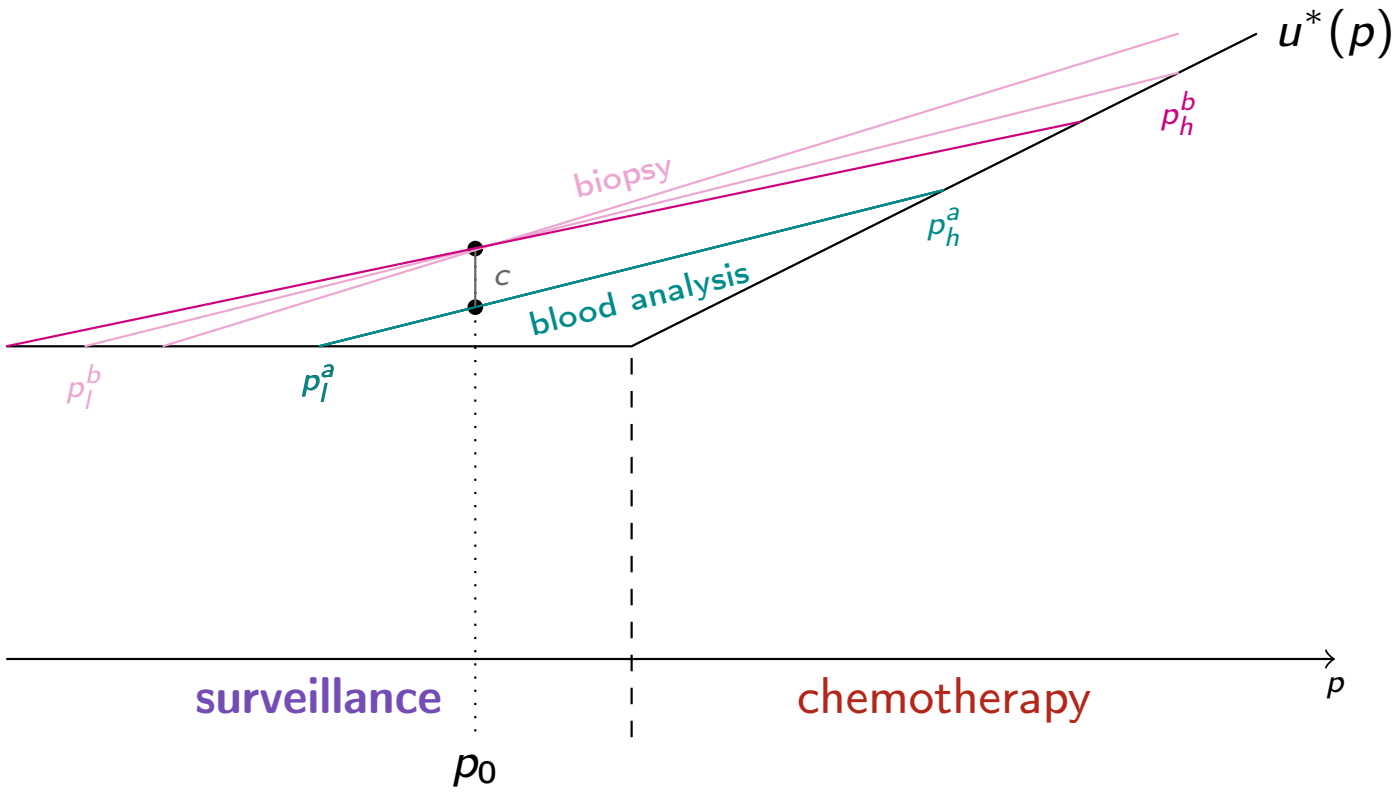
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Theorem

Suppose that $\mathbb{E}_{\mu(a_1)} p = \mathbb{E}_{\mu(b_1)} p$ *for all* a_1, b_1, *and*

- *actions with nontrivial continuation are connected in the adjacency graph, and*
- *every other action is adjacent to one of them.*

Then, X *is incentivizable if and only if it is equivalent to*

$$X(a_1, p) = s\, u^*(a_1, p) + d(p) \quad \text{for } s \in \{0, 1\} \text{ and affine } d : \Delta\Theta \longrightarrow \mathbb{R}.$$

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"What are your expected payoffs?"

"How likely will you believe that chemotherapy is optimal?" $1\{p > p^*\}$ *is not affine in* p.

Dynamic problems: Period 1

Actions affect states

Theorem

Suppose that the period 1 problem is nontrivial. Then, X is incentivizable if and only if it is equivalent to

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"What are your expected payoffs?"

Any question about learning, or about the states.

Dynamic problems: Period 2

Student.

Example: Student

- Student decides whether to write an *optional* test with a single True-False question.
- If he stays out, his grade is $h = 0.8$ (80%).
- If he writes the test, its weight is 25%.
- The decision is made before the Student sees the test.
- The test reveals information about its difficulty.

- $\Theta = \{0, 1\}$
- $A_1 = \{0, 1\}$, $A_2 = \{0, 1\}$
- test score:

$$P(a_2, \theta) = 1\{a_2 = \theta\}$$

- payoffs:

$$u(a_1, a_2, \theta) = h + \frac{1}{4} a_1 (P(a_2, \theta) - h)$$

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"What is the probability that the correct answer is True?"

$$X(a_1, a_2, \theta) = 1\{\theta = 1\}.$$

"What is your expected grade?"

$$X(a_1, a_2, \theta) = u(a_1, a_2, \theta).$$

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- Question $X(a_1, a_2, \theta)$ is *incentivizable in period 2* in dynamic decision problem (A_1, A_2, u) if there exists nondistortionary expansion $(A_1, A_2 \times \mathbb{R}, v)$ such that

$$\begin{aligned} \text{no distortions in } t = 1\text{: } & \mathbb{B}_1^v(\mu) = \mathbb{B}_1^u(\mu), \\ \text{no distortions in } t = 2\text{: } & \mathbb{B}_2^v(a_1, p) \subseteq \mathbb{B}_2^u(a_1, p) \times \mathbb{R} \\ \text{truthful reporting: } & \mathbb{B}_2^v(a_1, p) \subseteq \{(a_2, r) : r = \mathbb{E}_p X(a_1, a_2, \theta)\} \end{aligned}$$

  - $r \in \mathbb{R}$ - period 2 report of the expectation of a variable $X(a_1, a_2, \theta)$
- Incentivization in period 2 increases the value of information, which may distort behavior in period 1.

Example: Student

Value of information: a new argument

- So far: *value matching*, i.e., convex vs. affine value function of the reporting DM.
- If X is *incentivizable*, the first-period indifference condition must be preserved for every $\mu \in \Delta\Delta\Theta$:

$$\mathbb{B}_1^u(\mu) = \{a_1, b_1\} = \mathbb{B}_1^v(\mu).$$

- Hence, continuation payoffs

$$u^*(a_1, \cdot) - u^*(b_1, \cdot) \sim v^*(a_1, \cdot) - v^*(b_1, \cdot)$$

are collinear.

Example: Student

Value of information: a new argument

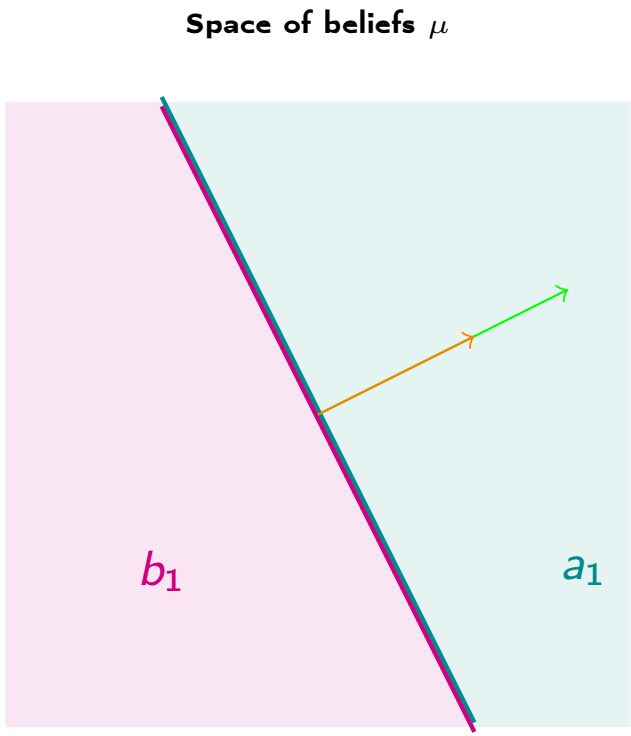
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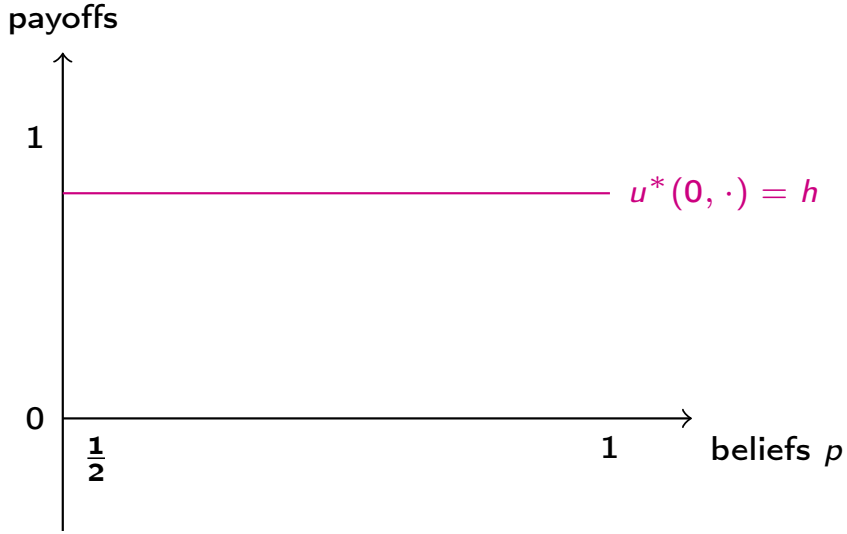
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Example: Student

Period 2 necessary conditions

- "What is your expected grade?" $X(a_1, a_2, \theta) = u(a_1, a_2, \theta)$.

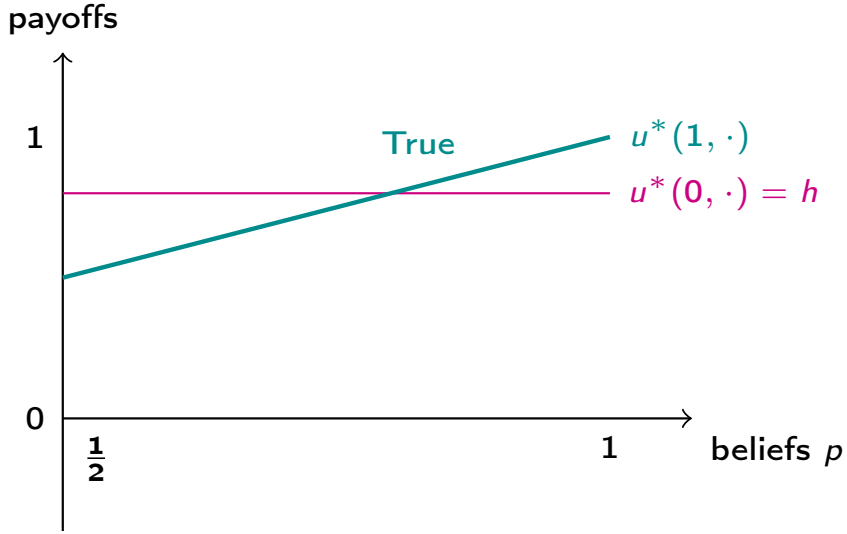


- $u^*(0, \cdot) = h$ is constant.
- $u^*(1, \cdot)$ is affine on $[\frac{1}{2}, 1]$: the optimal answer is True.
- $u^*(1, \cdot) - u^*(0, \cdot)$ is affine.

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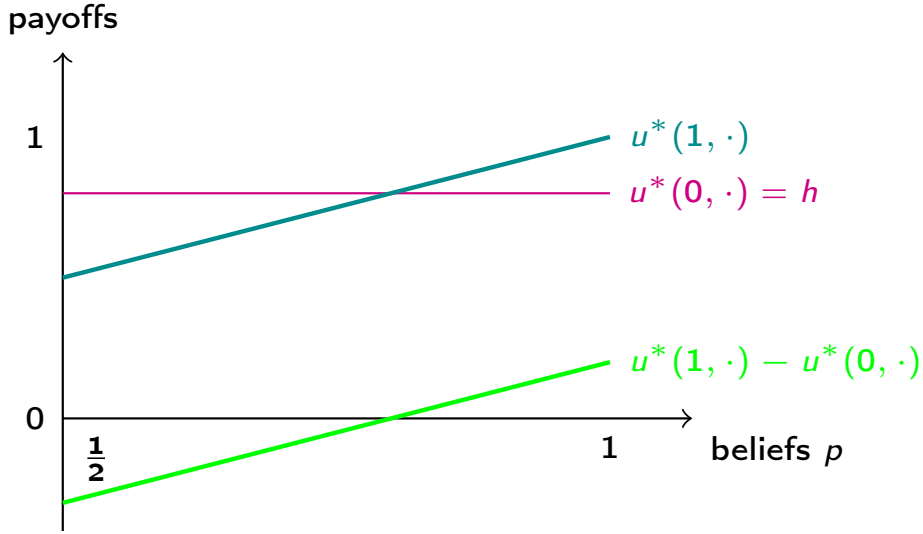


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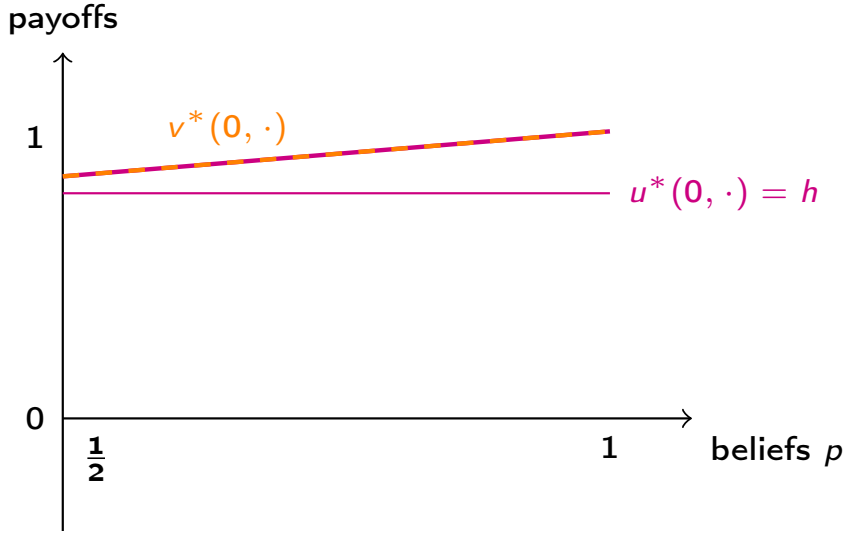


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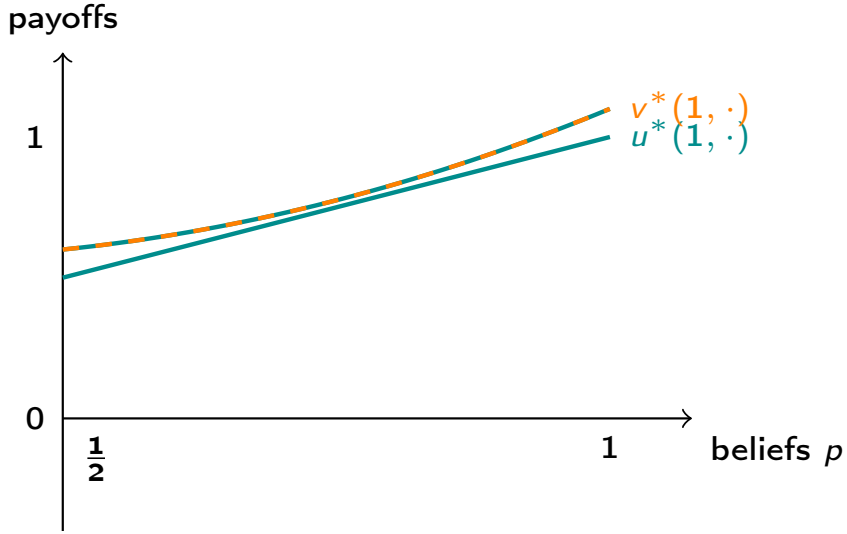


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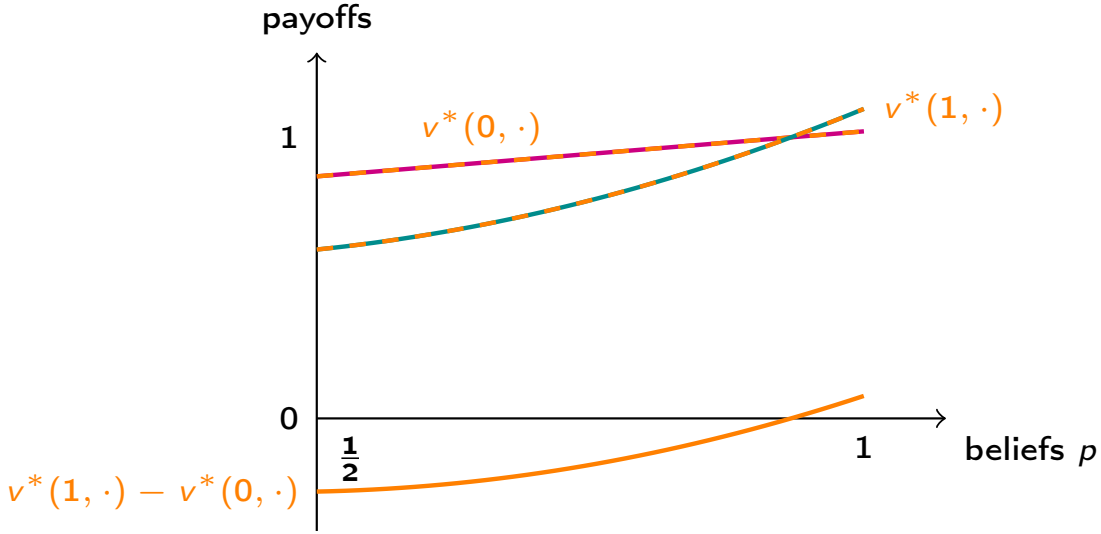


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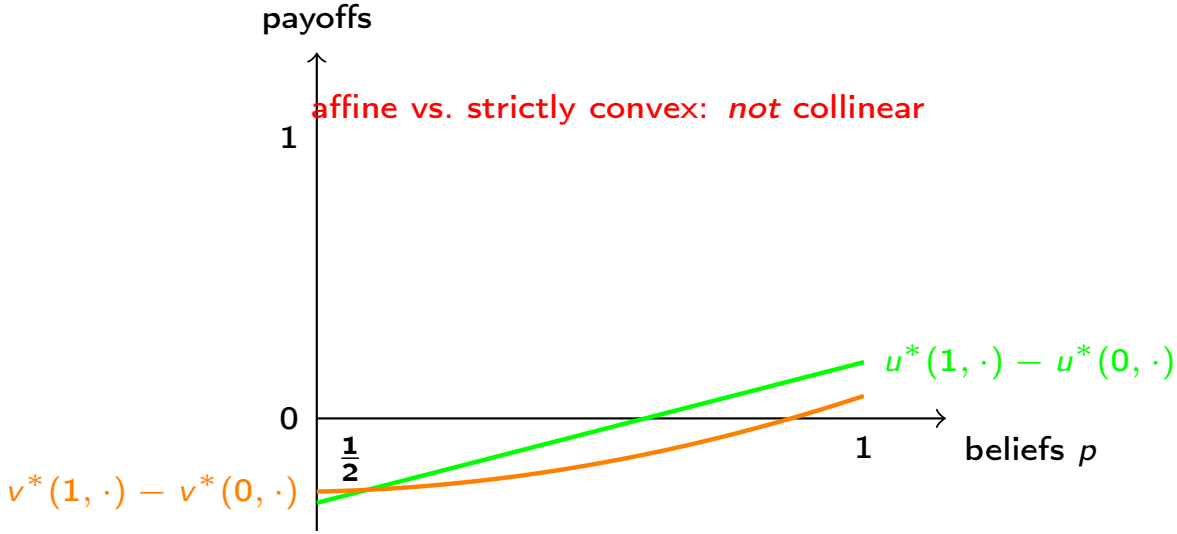


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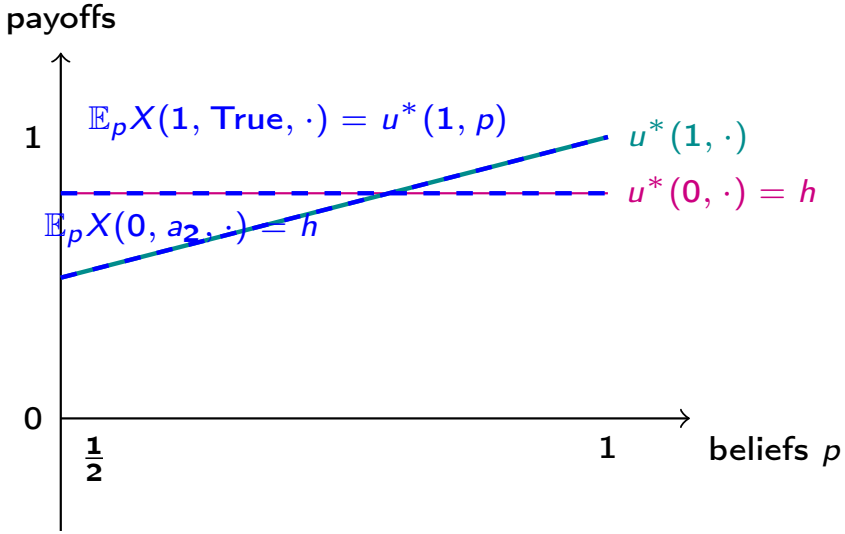


- No distortions in period 1 $\implies$ $v^*(1,\cdot) - v^*(0,\cdot)$ and $u^*(1,\cdot) - u^*(0,\cdot)$ are collinear: impossible.
- $\implies$ X is not incentivizable: incentivizing the revelation of information in period 2 *must* change the incentives in period 1.

Example: Student

Period 2 necessary conditions

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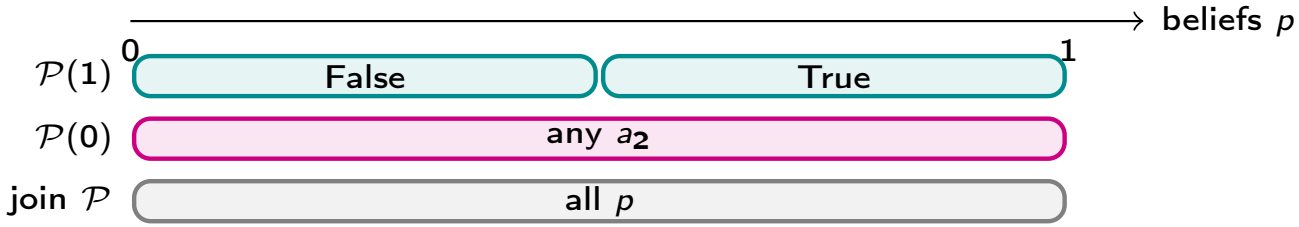


- What is the problem? On $[\frac{1}{2}, 1]$, $\mathbb{E}_p X(1, \text{True}, \cdot)$ changes while $\mathbb{E}_p X(0, a_2, \cdot)$ is constant.
- Incentivizable $X \implies X(a_1, a_2, \cdot) \sim X(b_1, b_2, \cdot)$ whenever a_2, b_2 are the continuation best responses after a_1, b_1 on a common interval of beliefs.

Dynamic problems: Period 2

Join of belief partitions

- Cells $P^u(a_1, a_2) = \{p : a_2 \in \mathbb{B}_2^u(a_1, p)\}$ of $\mathcal{P}(a_1)$. Overlapping cells after adjacent $a_1, b_1 \implies X(a_1, a_2, \cdot) \sim X(b_1, b_2, \cdot)$.
- *Join* $\mathcal{P}$: merge the cells across the overlaps. Incentivizable X is equivalent to a question that depends only on the cell of the join.

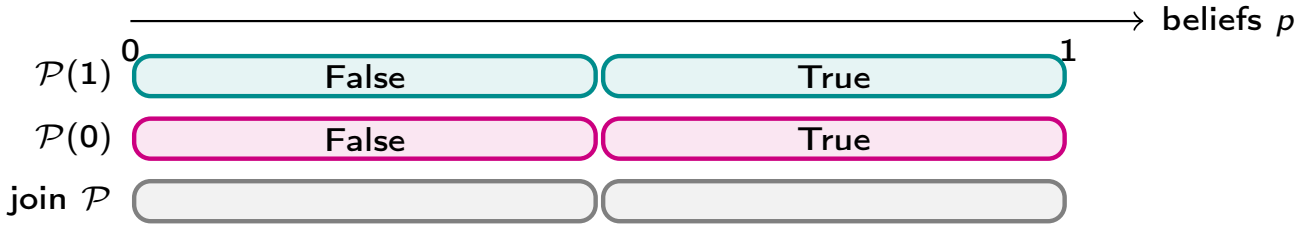


- Student: the join is trivial $\implies$ X is equivalent to an action-independent question.
- *Separable* payoffs, $u = u_1(a_1, \theta) + u_2(a_2, \theta)$: $\mathcal{P}(0) = \mathcal{P}(1)$ and the join is $\mathcal{P}(a_1)$ $\implies$ X depends on a_2 only.
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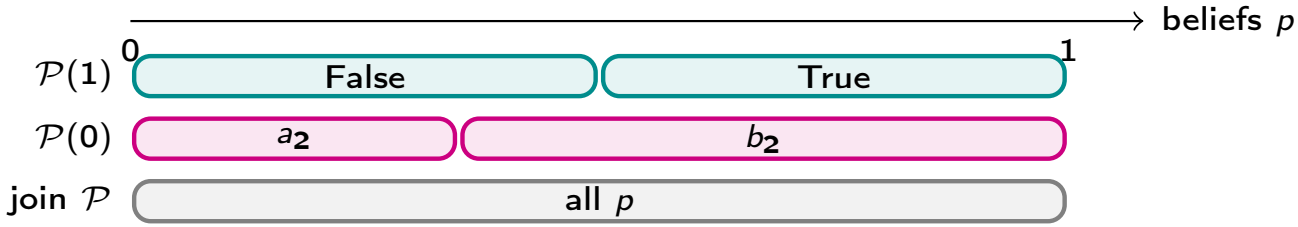


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- *Generic* payoffs: different cut points, the join is trivial $\implies$ X depends on θ only.

Theorem

Suppose that information is independent of actions, and

- *payoffs are essentially separable:* $\mathbb{B}_2^u(a_1, p) = \mathbb{B}_2^u(b_1, p)$ *for all essential* a_1, b_1*, and*
- *the period 2 problem satisfies the Pęski and Stewart (2025) conditions (e.g., complete adjacency graph).*

Then, X *is incentivizable if and only if it is equivalent to*

$$X(a_1, a_2, \theta) = s\, u(a_1^*, a_2, \theta) + d(\theta) \quad \text{for } s \in \{0, 1\} \text{ and } d \in \mathbb{R}^{\Theta}.$$

Any question about expected payoffs.

Theorem

Suppose that information is independent of actions, and

- *the period 1 problem is nontrivial, and*
- *payoffs are generic: no difference $u(a_1, a_2, \cdot) - u(a_1, b_2, \cdot)$ is a nonzero scalar multiple of $u(b_1, a_2', \cdot) - u(b_1, b_2', \cdot)$ for essential $a_1 \neq b_1$.*

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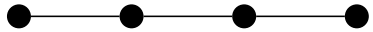
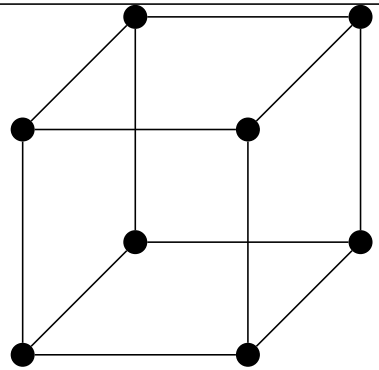
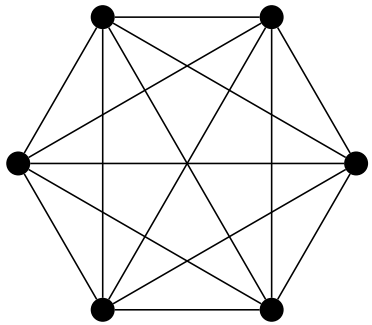
$$X(a_1, a_2, \theta) = d(\theta) \quad \text{for } d \in \mathbb{R}^{\Theta}.$$

Questions about the state only.

Any action-dependent question.

Conclusions

Incentivizable questions in static problems

	Tree	**Product problem**	**Complete graph**
Adjacency graph			
Examples	monotone problems, cognitive uncertainty (Enke and Graeber, 2023),	random problem selection (Azrieli et al., 2018), test with ≥ 3 questions	multiple choice question, prediction problems
Representation (necessary = sufficient)	Adjacency Lemma on each adjacency pair	$\sum_i s_i u_i(a_i, \theta_i) + d(\theta)$ where $s_i \in \mathbb{R}$ and equivalents*	$s\, u(a, \theta) + d(\theta)$ where $s \in \{0, 1\}$ and equivalents*

***Additional independence assumptions on payoffs**

Conclusions

Complete characterization

Incentivizable questions X:

	Action-independent information	Actions may affect the quality of information	Actions may affect the states
$t = 1$	$s\,u^*(a_1, p) + f(p)$ PS25 conditions	$s\,u^*(a_1, p) + d(p)$ nontrivial continuation	$s\,u^*(a_1, p)$ always
$t = 2$	$s\,u(a_1^*, a_2, \theta) + d(\theta)$ only if payoffs are separable	✗	✗

- $s \in \{0, 1\}$
- $f(p)$ is an arbitrary (sufficiently regular) function of beliefs
- $d(p)$ is affine in p